JONATHAN RUFUS SAMUEL

Systems Software Engineering | Research Engineer | Open-Source Developer | Systems & Cloud Infrastructure

EDUCATION

Vellore Institute of Technology *November, 2020 - August, 2024*
B.Tech Computer Science Engineering with Specialization in IoT GPA: 8.51 (3.4/4.0)

- Courses: Data Structures and Algorithms, Software Engineering, Computer Architecture, Microprocessors, Edge Networking, IoT, Fog & Edge Computing, Digital Signal Processing, Machine Learning, Information Security & Analysis

Research Experience - Student Research & Development Assistant *February, 2021 - August, 2024*

- Worked with Jaskaran Singh, Shreyash Mehrotra and Dr. Baiju B. V. to research the simulation challenges faced by IoT developers in the field of Fog and Edge Computing, and how to overcome such simulation challenges via open-source development. Published in proceedings of 2023 International Conference on Next Generation Electronics (NEleX). DOI: IEEE Conference Record # 59773, IEEEXplore ISBN NO: 979-8-3503-1908-8
- Worked with Shivansh Sahai and Dr. Swarnlatha P to create Automated Mp3 Tag Editor, a tool which scrapes metadata of music tracks from various sources, and classifies them via vector-based indexing, whilst automating the entire process to smoothly manage music metadata. Published as a chapter in Handbook of Research on Deep Learning Techniques for Cloud-Based Industrial IoT by IGI Global. DOI: 10.4018/978-1-6684-8098-4.ch012. Developed it further into an open-source **Rust based Tauri Desktop Application** to support extended music playback features.
- Worked with Dr. Yokesh Babu to propose and simulate a Zigbee based Commuter Barrier system to prevent commuter accidents on railway platforms in India

Bishop Cotton Boys' School *June, 2005 - March, 2020*
ISC & ICSE (High School) ISC: 93.5% & ICSE: 91.2%

- Headed the Logical Reasoning and Mathematical Event, "EventX", in Bangalore's biggest inter-school Science and IT Fest, Synchronize. (2019-2020)
- Member of the School Robotics Team for a duration of 4 years (20012-2016), placed in the Top 10 teams (Ranked 5th) in the All-India Robotics Olympiad (2015) out of 150 teams from all over India.

PROFESSIONAL EXPERIENCE

DevOps Engineer for Windows *Geneva, Switzerland*
CERN - European Organization for Nuclear Research **September, 2024 - Present**

- DevOps Engineer for Windows - Work on Systems Administration for Windows, DevOps Engineering for Sysadmin services and Software Development for supporting internal services.
- Tasked with Packaging, configuration, central provisioning and management of applications and general centralized Administration of CERN Windows server infrastructure and Windows end-point device park of over 10,000 devices.
- Aided the multi-year migration from in-house management software for over 10,000 machines to hybrid industry backed solutions - Microsoft Configuration Manager and Intune. Developed Package Automator (More information Below).
- Revamped and reverse engineered two 20+ year old applications: the Email Alert Web App and NICE Admin Review site, migrating them to .NET Core to support email alert delivery and local admin management across 10,000+ machines.
- Developed a fully configurable notifications service with a simple compiler to dynamically interpret custom configurations into code - Custom Notification template forms for administrators depending on internal DB constraints.
- **Tools Used:** C#, Powershell, Microsoft Configuration Manager (MECM), Intune, Azure, Entra, Active Directory, Python, React, Git, Gitlab, OpenShift, OpenStack, Docker, Kubernetes, MSSQL, Influx DB, Grafana.

Research & Development Experience - IT-CD-DPP Research Group @ CERN *September 2024 - Present*

- Spearheaded research and development of Package Automator - a Provisioning via code CI/CD tool that helps automatically create, test and deploy packages across both Intune and MECM. Cut down day to day tasks of team from over 7 days of work each month to 1 hour. Set to be presented as **Next-gen tools for the new era of Windows device management at CERN** at HEPiX Fall Workshop Conference held at the University of Nebraska, USA (October 2026)
- Worked with Siavas Firoozbakht (CERN) et al. in presenting a poster titled **'A new era for the management of 10,000 Windows devices at CERN'** at Chulalongkorn University in Bangkok, Thailand; detailing development of in-house tooling to support transition to MECM and Intune. DOI:10.13140/RG.2.2.27034.81605

- Worked with Siavas Firoozbakht (CERN) et al. in presenting **Windows device management at CERN: A new era** at ALPS CSCS in Lugano, detailing the transition of CERN's Windows infrastructure from its proprietary device management tool to Intune and MECM. Contributed by integrating PSADT standard for packing and testing applications, as well as standardizing the overall workflow among all three management providers. DOI:10.13140/RG.2.2.33745.70241

SDE Intern - Backend Engineering *Bengaluru, India*
Hyperface **January, 2024 - July, 2024**

- Led a team of 3 Application Engineers to complete company wide APIGEE updation and migration, spanning over 4 microservices within the Hyperface application suite.
- Wrote Python Scripts for migrating over 2 million bank records to our servers for further data processing and analysis.
- Designed and developed a centralized Cron Tracker Mechanism within our orchestrator service and developed the end-to-end functionality for AU Bank's Virtual Card Handling feature, whilst leading a team of 3 Application Engineers.
- Was tasked in mentoring and conducting Knowledge Transfer sessions for 5 Application Engineers based in our Jaipur office. Lead multiple projects with their assistance.
- Handled Information Security issues with Bank Dashboards, including encryption of insecure data fields, fixing enumeration issues, recreating security breaches via Burpsuite, and addition of code to preserve security.
- Developed an Employee Management System as part of training for Java Spring Boot Development. Worked on CRUD operations development, authentication and authorization, testing and migration to Apache Groovy.
- **Tools Used:** Java, Groovy, Spring Boot, Spring Security, Apache Kafka, Postman, MySQL, Python Scripting, Burpsuite.

PROJECTS

Dijkstra, Bengaluru/Remote — GitHub/Dijkstra-Edu *February, 2023 - present*

- Started an open-source organization back in University. A collection of services built to cater towards preparing students for the job market by actively evaluating their profiles and giving personalized feedback based on their goals.
- Lead Software Architect; Developed and currently maintaining an intricate system of services, with over 20 repositories written in various languages and frameworks, from Next.js, TypeScript, Python, FastAPI, Rust & Axum, Go & Gin and Java Spring Boot. This is to expose students to various technologies and frameworks.
- Led design and student-run development of various sub-projects including a latex resume builder, a blog mechanism for students, a certificate generator based on their proof of work, a GitHub statistics aggregator and a custom Task Tracker.
- Actively mentored over 50 students on software development, early career orientation and best practices, as well as maintaining documentation, software infrastructure and deployment lifecycles.
- Designed and actively developed an experimental multi-goal programming language called **Dijk**, to help teach and dissect compiler design, whilst trying to build a microservice in the DIjkstra application suite using the language.

Chronicles, Geneva/Santiago — GitHub/Chronicles-Network-Manager *October 2025 - January, 2026*

- Developed a decentralized local phone book/people database system to help remember details and maintain professional and personal relationships. Presented as a lightning talk at CSC Santiago, 2026. **DOI:** 10.13140/RG.2.2.32002.44489

OTHER ACTIVITIES & ACHIEVEMENTS

- Completed Microsoft Azure Administrator Associate (AZ-104) & Microsoft Azure Fundamentals (AZ-900) Certifications.
- Former Beta Level Student Learn Ambassador at Microsoft (MLSA Program). Designed custom Azure Certification programs for students at VIT Vellore as part of local organization of Cloud Skills Challenge.
- Completed Mathematics for Machine Learning Specialization offered by Imperial College London & Completed Accelerated Computer Science Fundamentals Specialization offered by University of Illinois (Goldman Sachs ECHP-23 Program)
- CERN Guide for the Atlas Experiment, Anti-matter Factory (Anti-Proton Decelerator) and the CERN Data Center.
- Attended the CERN school of computing 2026 held in Latin America, Santiago Chile at the Pontificia Universidad Católica de Chile (UC). Earned a Diploma in Physics Computing, Software Engineering and Data Technologies
- Former Senior Core Member at IEEE-TEMS, Mentor to over 30 Junior students.
- Open Source Contributor at YAFS - Yet Another Fog Simulator, AppFlowy, CERN Root, Apprentice and Dijkstra.

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